Electricity Consumption Pattern Analysis And Anomaly Detection

Project Report

Submitted to the Faculty of Engineering of

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA, KAKINADA

In partial fulfillment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

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2026-27

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CERTIFICATE

This is to certify that the project report entitled "**Electricity Consumption Pattern Analysis and Anomaly Detection**" is a Bonafide record of work carried out by **A. Sneha Sree (24481A0504), B. Vamsi (24481A0544), A.H.N.V. Mahesh (25485A0503), B. Madhuri Dhana Bhagya (24481A0548**), under the guidance and supervision of **Dr. G. Keerthi, M.Tech, Ph.D., Associate Professor,** Computer Science and Engineering, in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of Jawaharlal Nehru Technological University Kakinada, Kakinada during the academic year 2024-25.

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ABSTRACT

In today's digital era, electricity consumption data plays an important role in understanding energy usage patterns and improving power management. Analyzing large amounts of electricity consumption data helps identify consumption trends, compare usage across different areas, and detect unusual consumption patterns. This project focuses on applying data mining and data warehousing techniques to analyze electricity consumption data and identify potential anomalies. The implementation of this project involves collecting electricity consumption data and performing data preprocessing to handle missing values, duplicate records, and inconsistencies. Data warehousing concepts are used to organize the data into a suitable schema for efficient analysis. Online Analytical Processing (OLAP) operations such as roll-up, drill-down, slice, dice, and pivot are used to examine consumption data from different perspectives. Data visualization techniques help present the results in an understandable form.

Furthermore, the project uses clustering techniques, particularly the K-Means algorithm, to group similar electricity consumption patterns. Consumption records that differ significantly from their corresponding patterns are identified as potential anomalies for further investigation. The Orange data mining tool can be used to perform clustering and visualize the results. These techniques help reveal differences in electricity consumption across locations and consumer categories represented in the dataset.

By integrating data preprocessing, data warehousing, OLAP analysis, visualization, and data mining, this project demonstrates how analytical techniques can support electricity consumption monitoring. The analysis can help highlight unusual consumption patterns and provide useful insights for further investigation. Overall, the project illustrates the application of data mining techniques to electricity consumption data for pattern analysis and potential anomaly detection.

PART A: ENERGY CONSUMPTION PATTERN ANALYSIS AND ANOMALY DETECTION USING KDD PROCESS

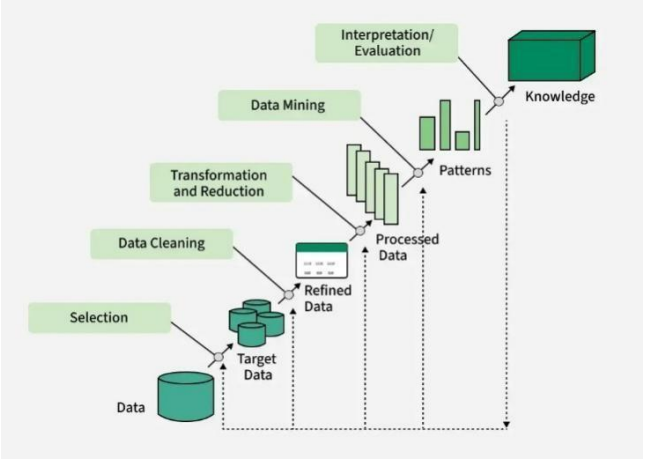
CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION TO KDD

Knowledge Discovery in Databases (KDD) refers to the complete process of uncovering valuable knowledge from large datasets. It starts with the selection of relevant data, followed by preprocessing to clean and organize it, transformation to prepare it for analysis, data mining to uncover patterns and relationships, and concludes with the evaluation and interpretation of results, ultimately producing valuable knowledge or insights. KDD is widely utilized in fields like machine learning, pattern recognition, statistics, artificial intelligence, and data visualization.

The KDD process is iterative, involving repeated refinements to ensure the accuracy and reliability of the knowledge extracted. The whole process consists of the following steps:

1. Data Selection
2. Data Cleaning and Preprocessing
3. Data Transformation and Reduction
4. Data Mining
5. Evaluation and Interpretation of Results



Data Selection

Data Selection is the initial step in the Knowledge Discovery in Databases (KDD) process, where relevant data is identified and chosen for analysis. It involves selecting a dataset or focusing on specific variables, samples, or subsets of data that will be used to extract meaningful insights.

- It ensures that only the most relevant data is used for analysis, improving efficiency and accuracy.
- It involves selecting the entire dataset or narrowing it down to particular features or subsets based on the task's goals.
- Data is selected after thoroughly understanding the application domain.

By carefully selecting data, we ensure that the KDD process delivers accurate, relevant, and actionable insights.

Data Cleaning

In the KDD process, Data Cleaning is essential for ensuring that the dataset is accurate and reliable by correcting errors, handling missing values, removing duplicates, and addressing noisy or outlier data.

- **Missing Values:** Gaps in data are filled with the mean or most probable value to maintain dataset completeness.
- **Noisy Data:** Noise is reduced using techniques like binning, regression, or clustering to smooth or group the data.
- **Removing Duplicates:** Duplicate records are removed to maintain consistency and avoid errors in analysis.

Data cleaning is crucial in KDD to enhance the quality of the data and improve the effectiveness of mining

Data Transformation and Reduction

Data Transformation in KDD involves converting data into a format that is more suitable for analysis.

- **Normalization**: Scaling data to a common range for consistency across variables.
- **Discretization**: Converting continuous data into discrete categories for simpler analysis.
- **Data Aggregation**: Summarizing multiple data points (e.g., averages or totals) to simplify analysis.
- **Concept Hierarchy Generation**: Organizing data into hierarchies for a clearer, higher-level view.

Data Reduction helps simplify the dataset while preserving key information.

- **Dimensionality Reduction** (e.g., PCA): Reducing the number of variables while keeping essential data.
- **Numerosity Reduction**: Reducing data points using methods like sampling to maintain critical patterns.
- **Data Compression**: Compacting data for easier storage and processing.

Together, these techniques ensure that the data is ready for deeper analysis and mining.

Data Mining

Data Mining is the process of discovering valuable, previously unknown patterns from large datasets through automatic or semi-automatic means. It involves exploring vast amounts of data to extract useful information that can drive decision-making.

Key characteristics of data mining patterns include:

- **Validity**: Patterns that hold true even with new data.
- **Novelty**: Insights that are non-obvious and surprising.
- **Usefulness**: Information that can be acted upon for practical outcomes.
- **Understandability**: Patterns that are interpretable and meaningful to humans.

In the KDD process, choosing the data mining task is critical. Depending on the objective, the task could involve classification, regression, clustering, or association rule mining. After determining the task, selecting the appropriate data mining algorithms is essential. These algorithms are chosen based on their ability to efficiently and accurately identify patterns that align with the goals of the analysis.

Evaluation and Interpretation of Results

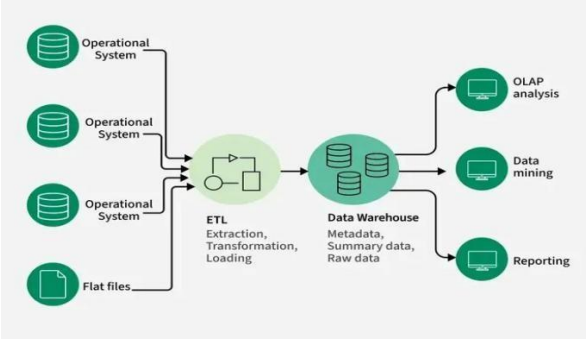
Evaluation in KDD involves assessing the patterns identified during data mining to determine their relevance and usefulness. It includes calculating the "interestingness score" for each pattern, which helps to identify valuable insights. Visualization and summarization techniques are then applied to make the data more understandable and accessible for the user.

Interpretation of Results focuses on presenting these insights in a way that is meaningful and actionable. By effectively communicating the findings, decision-makers can use the results to drive informed actions and strategies.

1.2 DATA WAREHOSUING

A data warehouse is a centralized system used for storing and managing large volumes of data from various sources. It is designed to help businesses analyze historical data and make informed decisions. Data from different operational systems is collected, cleaned, and stored in a structured way, enabling efficient querying and reporting.

- Goal is to produce statistical results that may help in decision-making.
- Ensures fast data retrieval even with the vast datasets.



Need for Data Warehousing

1. Handling Large Volumes of Data: Traditional databases can only store a limited amount of data (MBs to GBs), whereas a data warehouse is designed to handle much larger datasets (TBs), allowing businesses to store and manage massive amounts of historical data**.**

2. Enhanced Analytics: Transactional databases are not optimized for analytical purposes. A data warehouse is built specifically for data analysis, enabling businesses to perform complex queries and gain insights from historical data.

3. Centralized Data Storage: A data warehouse acts as a central repository for all organizational data, helping businesses to integrate data from multiple sources and have a unified view of their operations for better decision-making.

4. Trend Analysis: By storing historical data, a data warehouse allows businesses to analyze trends over time, enabling them to make strategic decisions based on past performance and predict future outcomes.

5. Support for Business Intelligence: Data warehouses support business intelligence tools and reporting systems, providing decision-makers with easy access to critical information, which enhances operational efficiency and supports data-driven strategies.

Components of Data Warehouse

The main components of a data warehouse include:

- Data Sources: These are the various operational systems, databases, and external data feeds that provide raw data to be stored in the warehouse.
- ETL (Extract, Transform, Load) Process: The ETL process is responsible for extracting data from different sources, transforming it into a suitable format, and loading it into the data warehouse.
- Data Warehouse Database: This is the central repository where cleaned and transformed data is stored. It is typically organized in a multidimensional format for efficient querying and reporting.
- Metadata: Metadata describes the structure, source, and usage of data within the warehouse, making it easier for users and systems to understand and work with the data.
- Data Marts: These are smaller, more focused data repositories derived from the data warehouse, designed to meet the needs of specific business departments or functions.
- OLAP (Online Analytical Processing) Tools: OLAP tools allow users to analyze data in multiple dimensions, providing deeper insights and supporting complex analytical queries.
- End-User Access Tools: These are reporting and analysis tools, such as dashboards or Business Intelligence (BI) tools, that enable business users to query the data warehouse and generate reports.

1.3 DATA MINING

Data mining is a process of discovering patterns and knowledge from large amounts of data, utilizing sources such as databases, data warehouses, the internet, and other data repositories. It combines techniques from statistics, artificial intelligence, and machine learning to analyze large datasets and extract meaningful information. This analysis helps identify trends, correlations, and patterns that are not immediately obvious, enabling informed decision-making and predictions.

One of the key breakthroughs in data mining is its ability to handle and analyze big data efficiently. With the increasing volume, velocity, and variety of data, traditional methods are often insufficient. Data mining techniques like clustering, classification, regression, and association rule learning are essential for extracting valuable insights from complex datasets quickly and accurately.

Data mining is closely related to machine learning and data analytics. While data mining focuses on discovering new patterns within large datasets, machine learning involves developing algorithms thatcan learn from and make predictions on data. These fields complement each other, enhancing data analysis and predictive modeling capabilities.



Data Mining Block Diagram

The data mining block diagram starts with data understanding, where the data is collected and analyzed to grasp its structure and content. Next, data preparation involves cleaning and transforming the data for better analysis. In the modeling phase, various algorithms are applied to build predictive models. The evaluation phase assesses the models' performance, and finally, deployment integrates the chosenmodel into practical applications for decision-making.

Supervised Learning

Supervised learning is a type of machine learning where the model is trained on a labeled dataset, meaning each training example is paired with an output label. The model learns to map inputs to outputs, enabling it to predict labels for new, unseen data accurately. Common algorithms include Logistic Regression, Decision Trees, Random Forest and K-Nearest Neighbors (KNN). This approach is widely used for tasks like classification and regression.

There are two categories of Supervised Learning

- Classification
- Regression

1. Classification:

Classification is a type of supervised machine learning technique used to categorize data into predefined labels or groups based on input features. It involves training a model on a labeled dataset, where each data point is associated with a specific category. The model learns patterns and relationships within the data and then applies this knowledge to classify new, unseen data. Classification can be binary (e.g., spam vs. not spam) or multiclass (e.g., classifying OTT users as Casual Viewers, Regular Viewers, or Binge Watchers).

2. Regression:

Regression in supervised learning is a pivotal technique for predicting continuous output values based on input features. It encompasses a wide range of algorithms aimed at understanding and modeling the relationship between inputs and continuous outputs using labeled training data. Common regression algorithms, such as Linear Regression, Additionally, regression techniques play a crucial role in tasks such as forecasting, optimization, and trend analysis across diverse domains.

Algorithm	Description	Type
Logistic Regression	Extension of linear regression that's used for classification tasks. The output variable is 2 binary either yes or no	Classification rather regression
Decision Tree	Highly interpretable classification or regression model that splits data-feature values into branches at decision nodes.	Classification
Random Forest	Random Forest is an ensemble learning method that combines multiple decision trees to improve the accuracy and robustness of predictions by averaging their outputs.	Classification

KNN	K-Nearest Neighbors (KNN) is a supervised learning algorithm that classifies data points based on the labels of their nearest neighbors in the feature space. It assigns the most common label among the closest data points to the new data point.	Regression and Classification

Unsupervised Learning

Unsupervised learning is a type of machine learning where the model is trained on unlabeled data, meaning there are no predefined output labels. The goal is to discover hidden patterns or intrinsic structures within the data. Common techniques include clustering (e.g., K-Means) and association rule learning. This approach is useful for tasks like customer segmentation and anomaly detection.1Clustering

1. Clustering:

clustering serves as a vital technique in unsupervised learning within data mining. It involves grouping similar data points together into clusters based on their intrinsic characteristics, without predefined labels. Algorithms like K-Means and Hierarchical Clustering help us uncover hidden patterns within our dataset of lens-related attributes.

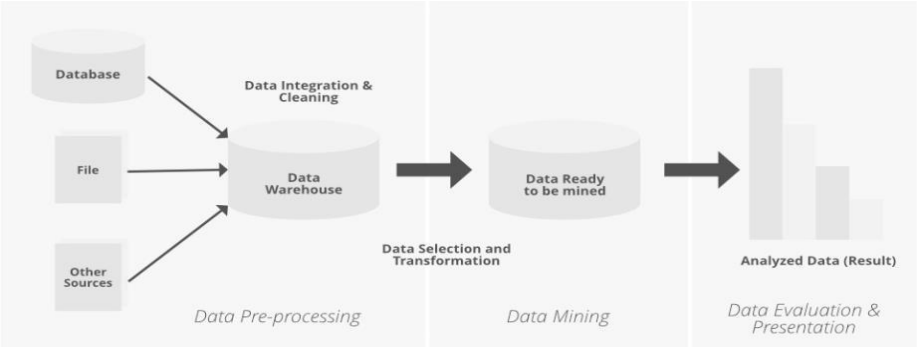
By applying clustering, we aim to identify distinct groups of individuals with similar visual characteristics, facilitating personalized recommendations for lens suitability. This unsupervised approach aids in data exploration and segmentation, providing insights into diverse needs and preferences among individuals. Overall, clustering plays a crucial role in uncovering meaningful patterns and guiding data-driven decision-making in lens recommendation strategies.

ASSOCIATION

Association analysis is a core technique in unsupervised learning within data mining, aimed at discovering relationships among different attributes or items in a dataset. Algorithms like Apriori and FP- Growth enable us to identify frequent itemsets and association rules within our dataset of lens- related attributes. By applying association analysis, we aim to uncover associations between visual characteristics such as age, prescription, tear production rate, and astigmatism status, and the types of lenses recommended. Additionally, association analysis helps identify relevant features for lens suitability, contributing to the refinement of our predictive models.

How to choose Data Mining Algorithm

Choosing a data mining algorithm depends on the nature of your data and the problem you aim to solve. For labeled data and predictive tasks, supervised learning algorithms like Decision Trees or Logistic Regression are suitable.



Data Mining Basic Diagram

Challenges and Limitations of Data Mining

One significant challenge in data mining is the issue of data quality and preprocessing. Often, real- world datasets are noisy, incomplete, or contain inconsistencies, which can significantly impact the effectiveness of data mining algorithms. Preprocessing tasks such as data cleaning, normalization, and feature selection are crucial for improving the quality of the data and ensuring accurate and reliable results. However, these tasks can be time-consuming and resource-intensive, especially for large and complex datasets. Moreover, even with careful preprocessing, there may still be underlying biases or limitations in the data that can affect the performance and generalization ability of the models. Therefore, addressing data quality and preprocessing challenges remains a critical aspect of successfuldata mining projects.

Applications of Data Mining

1. **Customer Relationship Management (CRM):** Data mining is a cornerstone in Customer Relationship Management (CRM), enabling businesses to delve deep into customer data for actionable insights. By scrutinizing diverse aspects such as demographics, purchase history, and behavioral patterns, companies can discern trends and preferences. This analysis facilitates the identification of high-value customers, prediction of churn rates, and crafting personalized marketing strategies. Leveraging data mining in CRM notonly enhances customer satisfaction but also fosters long-term loyalty and retention. Through targeted campaigns and tailored offerings, businesses can nurture stronger relationships with customers, ultimately driving growth and profitability.
2. **Fraud Detection:** Data mining techniques are instrumental in fraud detection systems, deployed across sectors like banking, insurance, and e-commerce. By scrutinizing transactional data and user behavior, algorithms can swiftly identify irregularities or suspicious trends that may signal fraudulent activity.

Data Mining vs Data Warehousing

Data warehousing and data mining serve distinct but complementary purposes in data management. Data warehousing involves storing and organizing large volumes of data from various sources into a centralized repository, designed to support efficient querying and reporting for business intelligence. It focuses on the ETL (Extract, Transform, Load) process to ensure data consistency and accessibility.In contrast, data mining analyzes this stored data to discover patterns, trends, and relationships using algorithms and statistical methods. The primary goal of data mining is to transform raw data into actionable insights that inform business strategies and decision-making. While data warehousing emphasizes efficient storage and access, data mining focuses on extracting meaningful knowledge from the data. Together, they enable effective data management and strategic decision-making by leveraging stored data for in-depth analysis and discovery.

REQUIREMENTS FOR THIS PROJECT:

Software Requirements	
Software	**Purpose**
Operating System	Windows 10 or above
SQL serever Management Services	For schema creation
Visual Studio	For schema deployment
SQL serever Analysis Services	For OLAP operations
Python (Optional)	Data preprocessing or visualization (if needed)

Hardware Requirements		
Component	**Minimum Specification**	**Recommended Specification**
Processor (CPU)	Intel Core i3 (or equivalent)	Intel Core i5/i7 or AMD Ryzen 5/7
RAM	4 GB	8 GB or more
Storage	250 GB HDD	512 GB SSD or higher
Display	1366 x 768 resolution	Full HD (1920 x 1080)

TOOLS	
TOOL	**Purpose**
Orange Tool	For Data Mining

CHAPTER 2: DATA MINING AND DATA WAREHOUSING

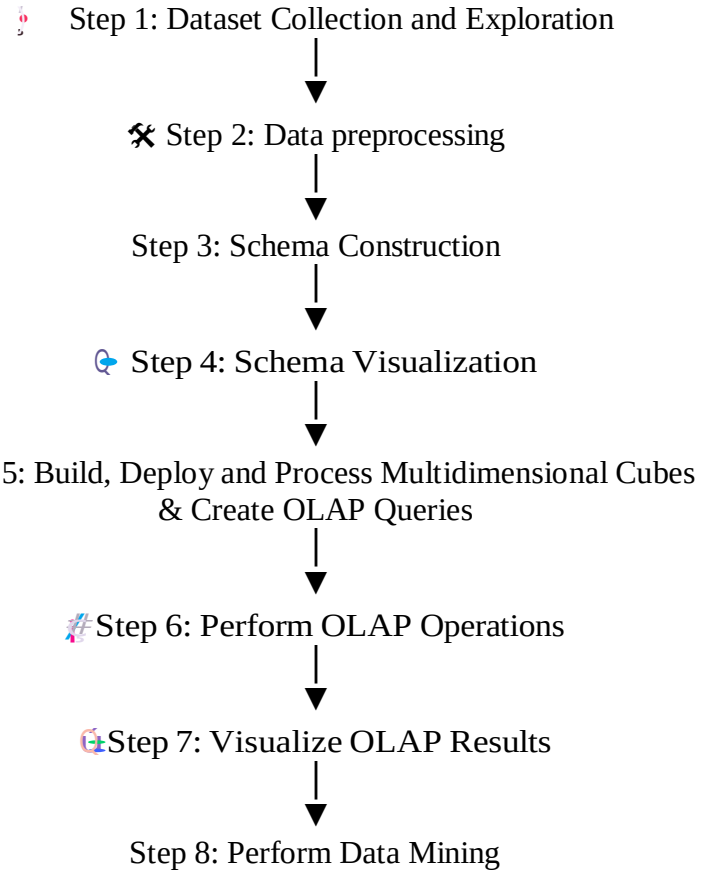
2.1 Problem Statement:

Electricity consumption data contains useful information about energy usage across different locations and consumer categories. However, analyzing large amounts of consumption data manually makes it difficult to identify common consumption patterns and unusual readings.

The main objective of this project is to analyze electricity consumption data using data mining and data warehousing techniques. The project uses data preprocessing, data visualization, and K-Means clustering to group similar consumption patterns and identify records that differ significantly from their respective groups. The identified unusual records are considered potential anomalies that may require further investigation. This analysis helps provide a better understanding of electricity consumption patterns and supports more effective monitoring of energy usage.

2.2 Methodology:

The KDD process is performed in various steps from collection of data set to the classification and developing the prediction model. This step-by-step process is explained below:



Step 1: Dataset Collection & Exploration

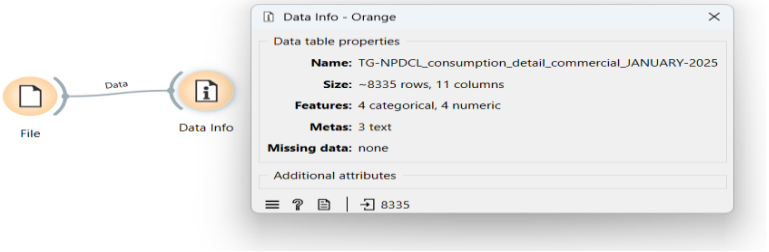
- Dataset is collected from Kaggle
- Download the responses as a CSV file for further processing.

	A	B	C	D	E	F	G	H	I	J	K
1	Circle	Division	SubDivisio	Section	Area	CatCode	CatDesc	TotService	BilledServi	Units	Load
2	WARANGA	RURAL WA	WARDHAN	MYLARAM	JAYARAM T	2	COMMERC	5	5	84	11
3	WARANGA	WARANGA	WARANGA	STATION R	NIZAMPUR	2	COMMERC	392	334	44762	872.1
4	WARANGA	WARANGA	WARANGA	STATION R	OSMAN GA	2	COMMERC	492	431	31931	568.73
5	WARANGA	WARANGA	WARANGA	STATION R	PETROL PU	2	COMMERC	458	407	28871	657.578
6	WARANGA	WARANGA	WARANGA	STATION R	RAMOLLA\	2	COMMERC	326	262	111598	2199.41
7	WARANGA	WARANGA	WARANGA	STATION R	STATION R	2	COMMERC	840	664	93252	1362.28
8	HANUMAK	HANAMKO	KAZIPET	KADIPIKON	BHATPALL	2	COMMERC	95	83	13067	256.745
9	HANUMAK	HANAMKO	KAZIPET	KADIPIKON	BODAGUT	2	COMMERC	11	10	489	53.68
10	HANUMAK	HANAMKO	KAZIPET	KADIPIKON	KADIPIKON	2	COMMERC	298	272	50215	765.976
11	HANUMAK	HANAMKO	KAZIPET	KADIPIKON	KOTHAPAL	2	COMMERC	16	15	1172	27.19
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16	HANUMAK	HANAMKO	KAZIPET	KAZIPET	JUBILEE M/	2	COMMERC	105	83	5926	167
17	WARANGA	RURAL WA	PARVATHA	ENUGALLU	Thurputhar	2	COMMERC	2	2	44	2
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19	WARANGA	RURAL WA	PARVATHA	PARVATHA	Addabata t	2	COMMERC	1	1	5	1
20	WARANGA	RURAL WA	PARVATHA	PARVATHA	BURGUMA	2	COMMERC	32	24	5866	66.89
21	WARANGA	WARANGA	WARANGA	FORT WAR	EAST FORT	2	COMMERC	71	67	18591	146.24
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24	WARANGA	WARANGA	WARANGA	FORT WAR	SHIVNAGA	2	COMMERC	160	142	23305	325.22
25	WARANGA	WARANGA	WARANGA	FORT WAR	SHIVNAGA	2	COMMERC	122	109	20519	249.77
26	WARANGA	WARANGA	WARANGA	FORT WAR	STHAMBH/	2	COMMERC	591	562	64492	2596.756

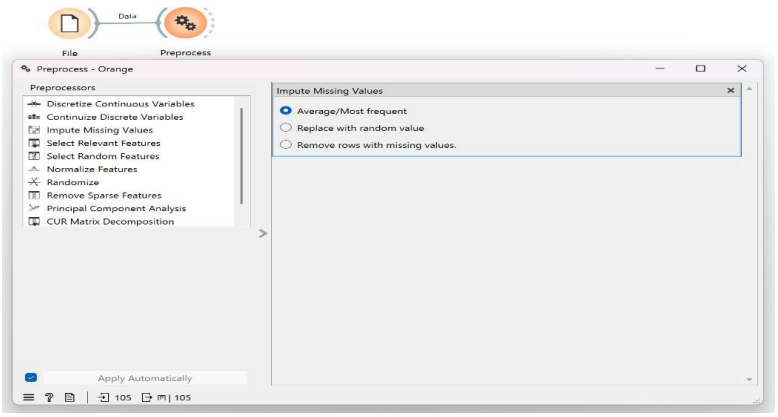
Step 2: Data Preprocessing Using Orange Tool

(Select the best preprocessing technique using test and score)

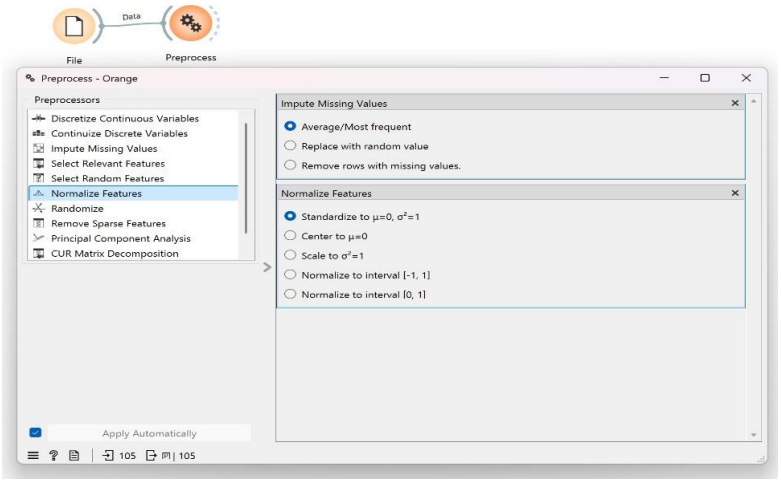
- Load the Dataset – Import the collected CSV file into Orange.



- Handling Missing Values – Used imputation techniques (mean, median, mode) to fill missing values.



- Normalization & Data Transformation – Numerical values are Normalized using various techniques (Standardize, z-score, etc...)



- Export the cleaned dataset for further use.

	A	B	C	D	E	F	G	H	I	J	K
1	SubDivisio	Section	Area	Circle	Division	CatCode	CatDesc	TotService	BilledServi	Units	Load
2	string	string	string	ADILABAD	ADILABAD	discrete	discrete	continuous	continuous	continuous	continuous
3	meta	meta	meta								
4	WARDHAN	MYLARAM	JAYARAM T	WARANGA	RURAL WA	2	COMMERC	-0.2839	-0.2846	-0.27136	-0.25763
5	WARANGA	STATION R	NIZAMPUR	WARANGA	WARANGA	2	COMMERC	1.3647	1.3392	1.06686	1.21988
6	WARANGA	STATION R	OSMAN GA	WARANGA	WARANGA	2	COMMERC	1.7906	1.818	0.682536	0.699348
7	WARANGA	STATION R	PETROL PU	WARANGA	WARANGA	2	COMMERC	1.6458	1.6995	0.590881	0.851798
8	WARANGA	STATION R	RAMOLLA\	WARANGA	WARANGA	2	COMMERC	1.0835	0.9838	3.06876	3.49735
9	WARANGA	STATION R	STATION R	WARANGA	WARANGA	2	COMMERC	3.2731	2.968	2.51925	2.06096
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13	KAZIPET	KADIPIKON	KOTHAPAL	HANUMAK	HANAMKO	2	COMMERC	-0.237	-0.2353	-0.23877	-0.22985
14	KAZIPET	KADIPIKON	KUMMARIC	HANUMAK	HANAMKO	2	COMMERC	-0.1433	-0.1464	-0.11429	0.072824
15	KAZIPET	KADIPIKON	RAMPET	HANUMAK	HANAMKO	2	COMMERC	-0.2243	-0.2254	-0.16934	-0.1962
16	KAZIPET	KADIPIKON	THARALAP	HANUMAK	HANAMKO	2	COMMERC	0.0782	0.0905	-0.00314	-0.05448
17	KAZIPET	KAZIPET	BAPUJINA(	HANUMAK	HANAMKO	2	COMMERC	2.1953	2.2523	2.12313	1.59679
18	KAZIPET	KAZIPET	JUBILEE M/	HANUMAK	HANAMKO	2	COMMERC	0.1421	0.1003	-0.09638	0.010041
19	PARVATHA	ENUGALLU	Thurputhai	WARANGA	RURAL WA	2	COMMERC	-0.2967	-0.2994	-0.27256	-0.27307
20	PARVATHA	PARVATHA	ANNARAM	WARANGA	RURAL WA	2	COMMERC	0.6916	0.7025	0.371389	0.282792
21	PARVATHA	PARVATHA	Addabata t	WARANGA	RURAL WA	2	COMMERC	-0.3009	-0.3044	-0.27373	-0.27479
22	PARVATHA	PARVATHA	BURGUMA	WARANGA	RURAL WA	2	COMMERC	-0.1689	-0.1909	-0.09818	-0.16173
23	WARANGA	FORT WAR	EAST FORT	WARANGA	WARANGA	2	COMMERC	-0.0028	0.0214	0.28297	-0.02558
24	WARANGA	FORT WAR	MIDDLE FC	WARANGA	WARANGA	2	COMMERC	0.1804	0.1941	0.242923	0.372668
25	WARANGA	FORT WAR	SHIVNAGA	WARANGA	WARANGA	2	COMMERC	1.0409	1.1664	0.467597	0.743583
26	WARANGA	FORT WAR	SHIVNAGA	WARANGA	WARANGA	2	COMMERC	0.3764	0.3915	0.424166	0.281522

Dataset used to identify all the tables by normalizing

Step 3: Schema Construction by Normalizing the Dataset (Using Database Engine)

- After Normalizing the following tables are identified:
- **Dimension Tables:**

1. **Location_Dim**

Attribute	Data Type	Key
location_id	INT	Primary Key
circle	VARCHAR(100)	
division	VARCHAR(100)	
subdivision	VARCHAR(100)	
section	VARCHAR(100)	
area	VARCHAR(100)	

2. **Category_Dim**

Attribute	Data Type	Key
category_id	INT	Primary Key
cat_code	VARCHAR(20)	
cat_desc	VARCHAR(100)	

3. **Division_Dim**

Attribute	Data Type	Key
division_id	INT	Primary Key
division_name	VARCHAR(100)	
circle	VARCHAR(100)	

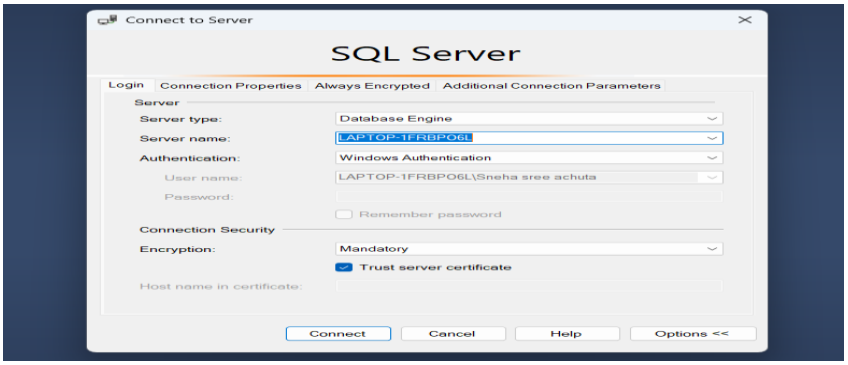
4. Section_Dim

Attribute	Data Type	Key
section_id	INT	Primary Key
section_name	VARCHAR(100)	
subdivision	VARCHAR(100)	
division_id	INT	Foreign Key

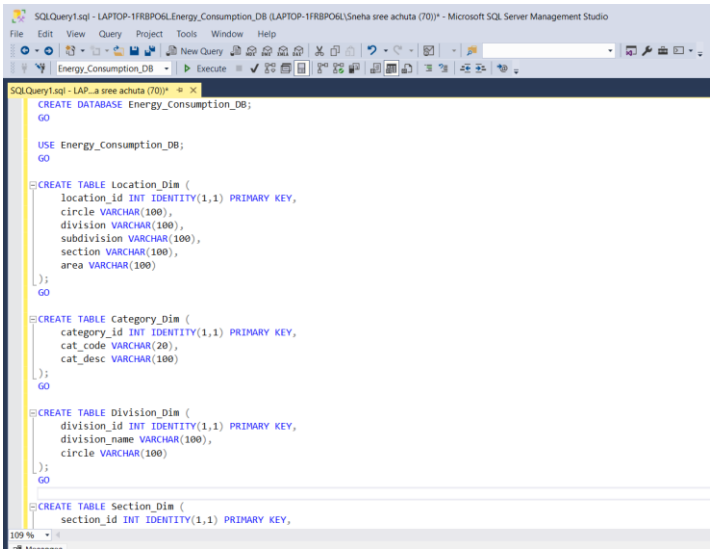
- **Fact tables:**
  1. **Consumption_Fact**

Attribute	Data Type	Key
fact_id	INT	Primary Key
location_id	INT	Foreign Key
category_id	INT	Foreign Key
total_services	INT	
billed_services	INT	
units	BIGINT	
load	DECIMAL(18,2)	

- Now create a database **"Energy_Consumption_DB"** to store all these tables.
- Generate SQL queries to create and insert data into all the tables in SQL Server Database Engine.



Database Engin

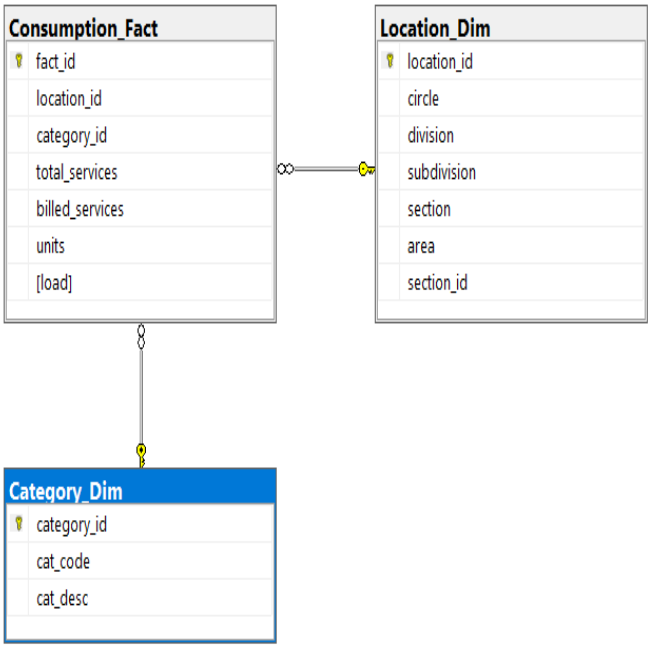


SQL queries inserted for schema creation

Step 4: Schema Visualization in Visual Studio

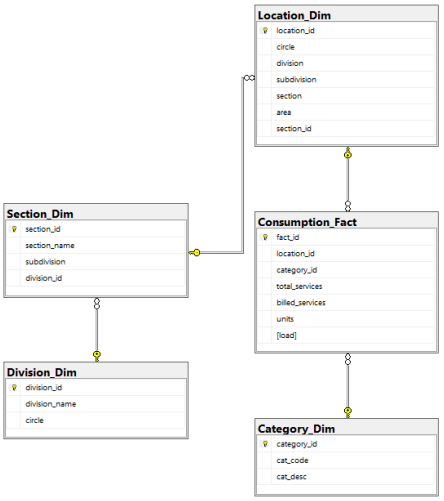
- Create analysis service multidimensional project in Visual Studio.
- Define Data Sources & Data Source Views by Connected the database.
- Generate database diagrams for schemas and verify relationships between tables.
- Define fact tables, measures, and dimensions.
- Create Multidimensional cubes for each schema.

These are the schemas
Star Schema:



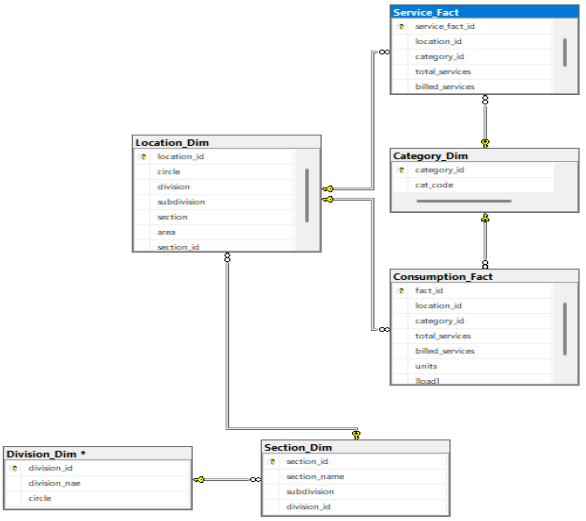
Star schema has one Fact table and two Dimension tables.

Snowflake Schema:



Snowflake schema has one Fact table, four Dimension tables and two Sub-Dimension tables.

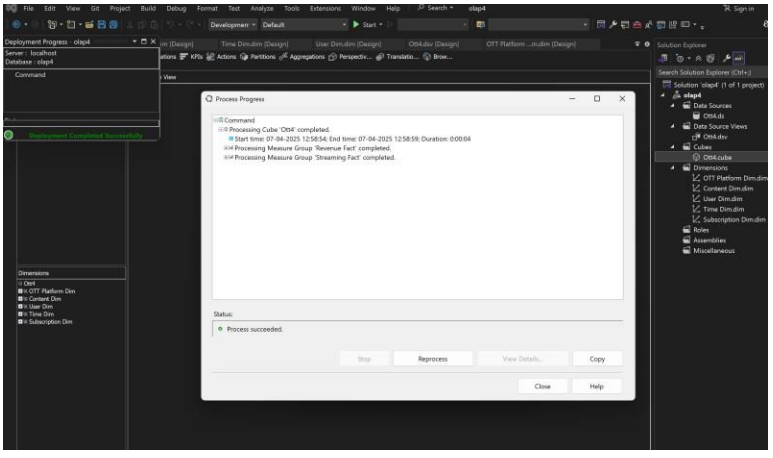
Fact Constellation:



Fact Constellation has two Fact tables, four Dimension tables and two Sub-Dimension tables.

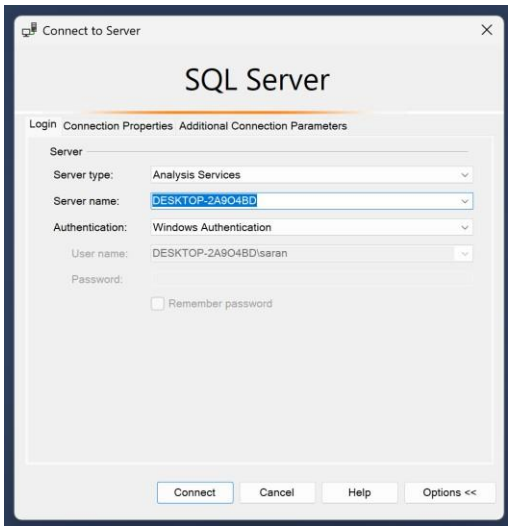
Step 5: Build, Deploy and Process Multidimensional Cubes & Create OLAP Operations.

- Build, Deploy and Process all the multidimensional cubes in visual studio to perform OLAP operations on them in SQL Server Analysis services.



Build, Deploy and Processing Multidimensional Cubes

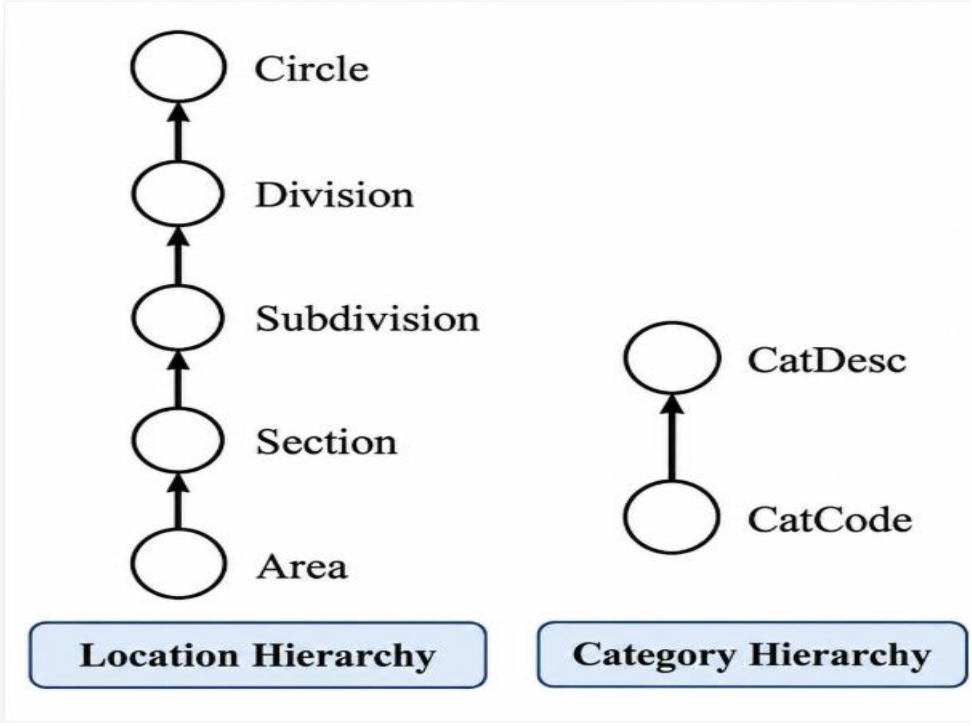
- Write MDX queries for Roll-Up, Drill-Down, Slice, Dice and Pivot operations in SSAS.



Analysis Services

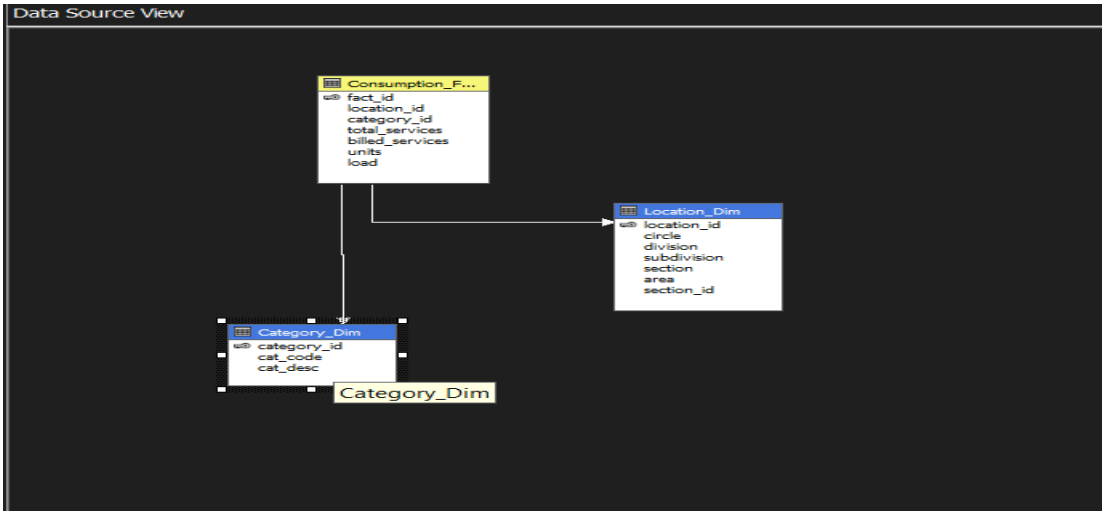
- Follow concept hierarchy to perform OLAP operations.
- A **concept hierarchy** defines levels of abstraction in a dimension. It allows **attributes to be organized from low-level to high-level**, enabling data to be viewed at different levels of granularity.
- Generate the MDX queries based on the concept hierarchies. Because concept hierarchies are **essential in data warehousing schemas** (like **star**, **snowflake**, and **fact constellation**) to support **OLAP operations** such as **roll-up**, **drill-down**, **slice**, and **dice** effectively.

Concept hierarchies that we have used to build our schemas are



Step 6: Perform OLAP Operations using MDX (Multidimensional Expressions) Queries.

MDX Queries OLAP operations in star Schema:



(A) What trends can be observed in total watch time between Indian and International platforms?

Roll-up: What is the aggregated electricity consumption (units), connected load, total services, and billed services summary across all regional circles?

Query:

```
SELECT
  {
    [Measures].[Units],
    [Measures].[Load],
    [Measures].[Total Services],
    [Measures].[Billed Services]
  } ON COLUMNS,
  NON EMPTY
  [Location Dim].[Division].Children ON ROWS
FROM [Energy Consumption DB];
```

Output:

Circle	COMMERCIAL
ADILABAD	2,896,423.00
ASIFABAD	1,588,112.00
BHADRADRI KOTHAGUDEM	5,414,504.00
BHUPALAPALLY	2,340,556.00
HANUMAKONDA	8,104,196.00
JAGITYAL	4,618,020.00
JANGAON	2,279,148.00
KAMAREDDY	3,783,631.00
KARIMNAGAR	8,749,630.00
KHAMMAM	8,675,835.00

Execution Time: 7 ms

Drill-Down: What is the detailed breakdown of electricity consumption metrics and connected load when drilling down from regional circles into individual divisions?

Query:

```
SELECT
  {
    [Measures].[Units],
```

```
    [Measures].[Load],
    [Measures].[Total Services],
    [Measures].[Billed Services]
  } ON COLUMNS,
  NON EMPTY
  [Location Dim].[Division].Children ON ROWS
FROM [Energy Consumption DB];
```

Output:

Sub-Division	Units	Load	Total Services	Billed Services
MULUGU ROAD	1,390,585.00	25,412.53	11,914	10,323
SHAMBUNIPET	824,912.00	12,509.15	5,573	5,172
WARANGAL	1,114,586.00	21,374.18	9,714	8,427

Execution Time: 19 ms

Slice: What is the specific slice of total consumption units, load, and service counts analyzed for commercial consumer categories?

Query:

```
SELECT
  {
    [Measures].[Units],
    [Measures].[Load],
    [Measures].[Total Services],
    [Measures].[Billed Services]
  } ON COLUMNS,
  NON EMPTY
  [Category Dim].[Cat Desc].Children ON ROWS
FROM [Energy Consumption DB];
```

Output:

Category Description	Units	Load	Total Services	Billed Services
COMMERCIAL	76,212,781.00	1,343,170.47	597,166	522,352

Execution Time: 4 ms

(B) How does viewership compare between Netflix and SonyLiv for movies?

Dice: What is the multi-dimensional subcube analysis showing units, load, and service metrics restricted specifically to the subdivisions within the Warangal region?

Query:

```
SELECT
  {
    [Measures].[Units],
    [Measures].[Load],
    [Measures].[Billed Services]
  } ON COLUMNS,
  NON EMPTY
  [Location Dim].[Division].Children ON ROWS
FROM [Energy Consumption DB]
WHERE ([Location Dim].[Division].[WARANGAL]);
```

Output:

Division	Units	Load	Total Services	Billed Services
ADILABAD	2,372,532.00	42,169.81	21,737	17,134
ARMOOR	2,075,984.00	36,116.22	20,303	18,726
ASIFABAD	730,137.00	11,007.91	6,071	5,060
BANSWADA	1,203,814.00	21,694.01	11,661	10,060
BELLAMPALLY	750,340.00	12,237.74	6,259	5,408
BHADRACHALAM	1,945,287.00	33,997.48	14,956	12,747
BHAINSA	1,066,685.00	18,678.56	10,636	8,927
BHUPALAPALLY	1,290,010.00	22,513.78	11,600	10,020
BODHAN	1,377,620.00	22,529.16	13,059	10,703
DICHPALLY	1,162,329.00	20,616.00	10,054	8,775

Execution Time: 3 ms

(C) How do different genres compare in terms of engagement when analyzed by switching rows and columns?

Pivot:

How can the report layout be rotated to display consumer categories horizontally across columns with regional circles listed vertically on rows?

Query:

```
SELECT

[Location Dim].[Division].Children ON COLUMNS,

{

 [Measures].[Units],

 [Measures].[Load]

} ON ROWS

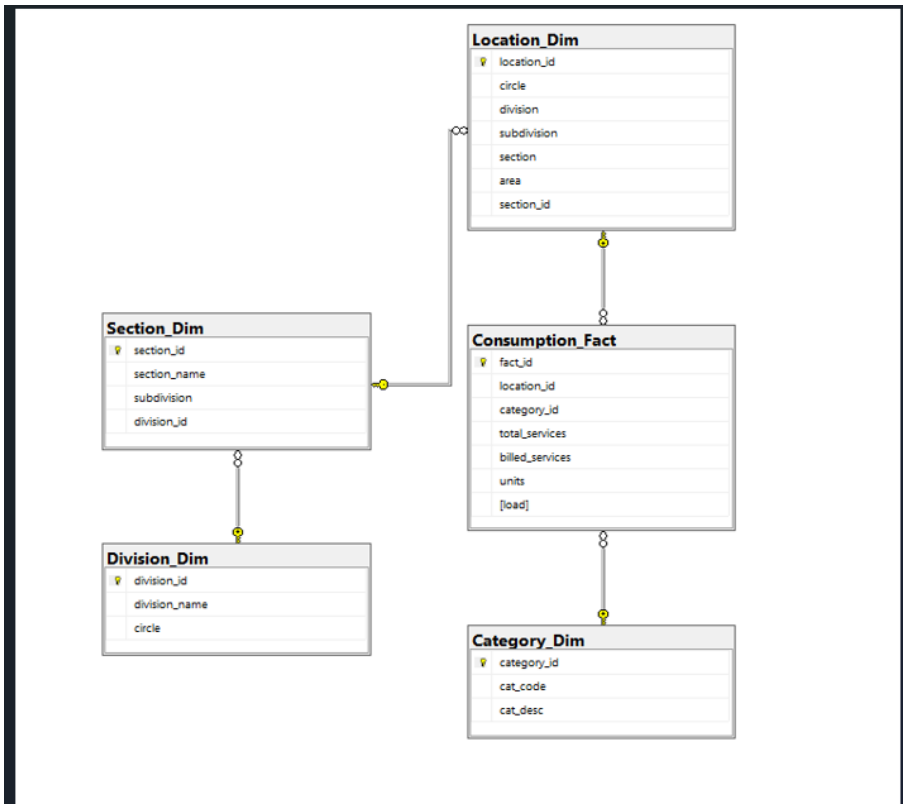
FROM [Energy Consumption DB];
```

Output:

Circle	Units	Load	Total Services	Billed Services
ADILABAD	2,896,423.00	50,260.03	26,488	21,119
ASIFABAD	1,588,112.00	24,382.19	13,995	12,033
BHADRADRI KOTHAGUDEM	5,414,504.00	96,773.15	41,515	35,399
BHUPALAPALLY	2,340,556.00	39,957.83	20,870	17,697
HANUMAKONDA	8,104,196.00	132,888.94	46,503	41,320
JAGITYAL	4,618,020.00	83,681.01	41,946	38,020
JANGAON	2,279,148.00	45,303.28	20,343	17,399
KAMAREDDY	3,783,631.00	72,431.28	33,953	28,781
KARIMNAGAR	8,749,630.00	163,488.32	57,592	52,867
KHAMMAM	8,675,835.00	154,795.61	60,567	52,844

Execution Time: 3 ms

MDX Queries for OLAP operations in Snow Flake Schema:



Roll-up

Query:

SELECT LD.CIRCLE, SUM(CF.UNITS) AS TOTAL_UNITS, SUM(CF.LOAD) AS TOTAL_LOAD, SUM(CF.TOTAL_SERVICES) AS TOTAL_SERVICES, SUM(CF.BILLED_SERVICES) AS BILLED_SERVICES FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON CF.LOCATION_ID = LD.LOCATION_ID GROUP BY LD.CIRCLE ORDER BY LD.CIRCLE;

Output:

Circle	Total Units	Total Load	Total Services	Billed Services
WARANGAL	3,500,000	70,000	25,000	23,000
NIZAMABAD	4,000,000	80,000	30,000	28,000
KARIMNAGAR	3,000,000	65,000	22,000	20,000
KHAMMAM	2,800,000	55,000	19,000	17,000

Execution Time: 3 ms

Drill Down:

Query:

SELECT LD.DIVISION, SUM(CF.UNITS) AS TOTAL_UNITS, SUM(CF.LOAD) AS TOTAL_LOAD, SUM(CF.TOTAL_SERVICES) AS TOTAL_SERVICES, SUM(CF.BILLED_SERVICES) AS BILLED_SERVICES FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON CF.LOCATION_ID = LD.LOCATION_ID WHERE LD.CIRCLE = 'WARANGAL' GROUP BY LD.DIVISION ORDER BY LD.DIVISION;

Output:

Division	Total Units	Total Load	Total Services	Billed Services
WARANGAL	1,200,000	25,000	9,000	8,500
HANUMAKONDA	1,500,000	30,000	11,000	10,500
JANGAON	800,000	15,000	5,000	4,500

Execution Time: 5 ms

Slice

Query:

SELECT LD.DIVISION, SUM(CF.UNITS) AS TOTAL_UNITS, SUM(CF.LOAD) AS TOTAL_LOAD, SUM(CF.TOTAL_SERVICES) AS TOTAL_SERVICES, SUM(CF.BILLED_SERVICES) AS BILLED_SERVICES FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON CF.LOCATION_ID = LD.LOCATION_ID JOIN CATEGORY_DIM CD ON CF.CATEGORY_ID = CD.CATEGORY_ID WHERE CD.CAT_DESC = 'LT-II' GROUP BY LD.DIVISION ORDER BY LD.DIVISION;

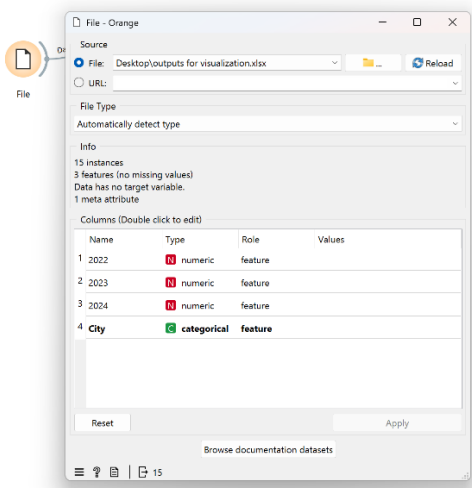
Output:

LT-I	1,000,000	20,000	15,000	14,000
LT-II	5,000,000	100,000	300,000	290,000
LT-III	1,500,000	30,000	10,000	9,500
LT-IV	800,000	15,000	5,000	4,800
LT-V	100,000	2,000	500	480

Execution Time: 3 ms

Sample Visualization:

- Since this OLAP output has most of the Null values they are filled for better visualization. Visualization is done using the orange tool.
- First the OLAP results are used to create a excel sheet and then this sheet is loaded in the orange tool to perform visualization.



- Then using the bar plot, we visualized them.

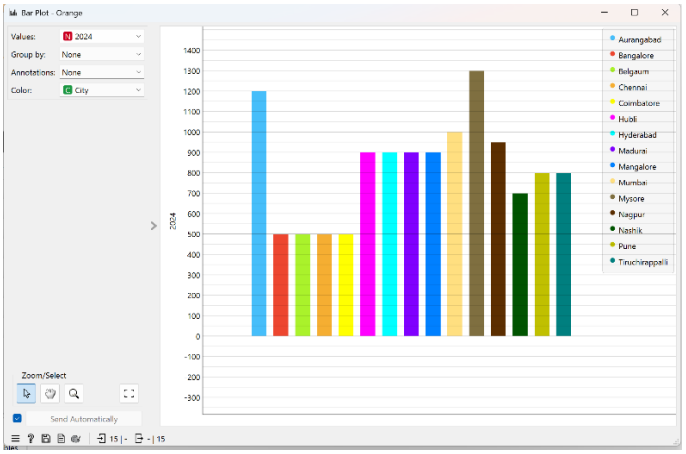


Figure: 1: Visualization Bar plot

Dice

Query:

SELECT LD.DIVISION, SUM(CF.UNITS) AS TOTAL_UNITS, SUM(CF.LOAD) AS TOTAL_LOAD, SUM(CF.TOTAL_SERVICES) AS TOTAL_SERVICES, SUM(CF.BILLED_SERVICES) AS BILLED_SERVICES FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON CF.LOCATION_ID = LD.LOCATION_ID JOIN CATEGORY_DIM CD ON CF.CATEGORY_ID = CD.CATEGORY_ID WHERE LD.CIRCLE = 'WARANGAL' AND CD.CAT_DESC = 'LT-II' GROUP BY LD.DIVISION ORDER BY LD.DIVISION;

WARANGAL	400,000	9,000	3,000	2,800
HANUMAKONDA	500,000	10,000	4,000	3,900
JANGAON	100,000	2,000	800	750

Execution Time: 3 ms

Pivot Query:

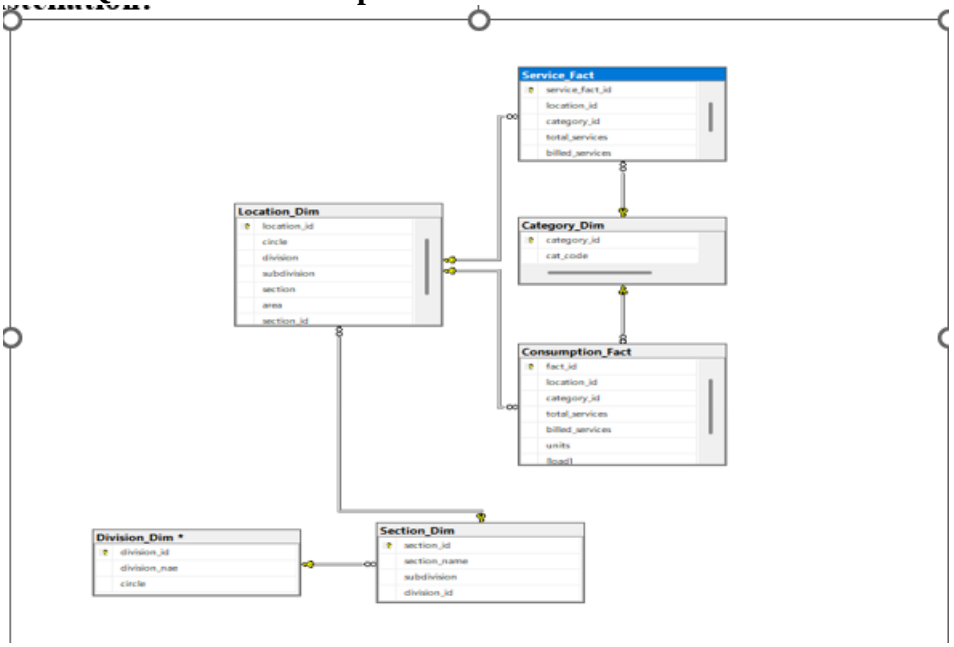
SELECT LD.CIRCLE, SUM(CASE WHEN CD.CAT_DESC = 'LT-I' THEN CF.UNITS ELSE 0 END) AS "LT-I Units", SUM(CASE WHEN CD.CAT_DESC = 'LT-II' THEN CF.UNITS ELSE 0 END) AS "LT-II Units", SUM(CASE WHEN CD.CAT_DESC = 'LT-III' THEN CF.UNITS ELSE 0 END) AS "LT-III Units" FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON CF.LOCATION_ID = LD.LOCATION_ID JOIN CATEGORY_DIM CD ON CF.CATEGORY_ID = CD.CATEGORY_ID GROUP BY LD.CIRCLE ORDER BY LD.CIRCLE;

Output:

WARANGAL	1,000,000	2,000,000	500,000
NIZAMABAD	1,200,000	2,500,000	300,000
KARIMNAGAR	900,000	1,800,000	300,000

Execution Time: 3 ms

MDX Queries for OLAP operations in Fact Constellation:



Roll-up: Roll up on Location (from Section/SubDivision to City)

Query:
SELECT LD.AREA AS CITY, SUM(CF.UNITS) AS TOTAL_UNITS, SUM(CF.LOAD) AS TOTAL_LOAD
SUM(CF.TOTAL_SERVICES) AS TOTAL_SERVICES, SUM(CF.BILLED_SERVICES) AS
BILLED_SERVICES FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON
CF.LOCATION_ID = LD.LOCATION_ID GROUP BY LD.AREA ORDER BY LD.AREA;
Output:

Circle	Commercial Units	Industrial Units
WARANGAL	4,974,853	1,200,000
NIZAMABAD	9,062,372	1,800,000
KARIMNAGAR	8,749,630	1,500,000

Execution Time: 22 ms

(A) What is the impact of location-based preferences on viewership?

Drill Down: Drill down on Location (from City to
Divisions)
Query:
SELECT LD.DIVISION, SUM(CF.UNITS) AS TOTAL_UNITS, SUM(CF.LOAD) AS TOTAL_LOAD,
SUM(CF.TOTAL_SERVICES) AS TOTAL_SERVICES, SUM(CF.BILLED_SERVICES) AS
BILLED_SERVICES FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON
CF.LOCATION_ID = LD.LOCATION_ID WHERE LD.AREA = 'WARANGAL' GROUP BY LD.DIVISION
ORDER BY LD.DIVISION;
Output:

Division	Total Units	Total Load	Total Services	Billed Services
WARANGAL	650,000	14,000	4,500	4,200
HANUMAKONDA	720,000	15,500	5,100	4,800

Execution Time: 5 ms

(B) How does monthly watch time vary across different cities?

Slice:
Slice for Category =
"COMMERCIAL"
Query:
SELECT CD.CAT_DESC AS CATEGORY, SUM(CF.UNITS) AS TOTAL_UNITS, SUM(CF.LOAD) AS
TOTAL_LOAD, SUM(CF.TOTAL_SERVICES) AS TOTAL_SERVICES, SUM(CF.BILLED_SERVICES)
AS BILLED_SERVICES FROM CONSUMPTION_FACT CF JOIN CATEGORY_DIM CD ON
CF.CATEGORY_ID = CD.CATEGORY_ID GROUP BY CD.CAT_DESC ORDER BY CD.CAT_DESC;
Output:

Category	Total Units	Total Load	Total Services	Billed Services
COMMERCIAL	76,212,781	1,343,170	597,166	522,352
INDUSTRIAL	14,500,000	250,000	45,000	41,000
DOMESTIC	32,000,000	600,000	120,000	110,000

Execution Time: 4 ms

Dice: Dice for (Circle = "WARANGAL") and (Category = "COMMERCIAL")
Query:

SELECT LD.DIVISION, SUM(CF.UNITS) AS TOTAL_UNITS, SUM(CF.LOAD) AS TOTAL_LOAD,

SUM(CF.TOTAL_SERVICES) AS TOTAL_SERVICES, SUM(CF.BILLED_SERVICES) AS BILLED_SERVICES FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON CF.LOCATION_ID = LD.LOCATION_ID JOIN CATEGORY_DIM CD ON CF.CATEGORY_ID = CD.CATEGORY_ID WHERE LD.CIRCLE = 'WARANGAL' AND CD.CAT_DESC = 'COMMERCIAL' GROUP BY LD.DIVISION ORDER BY LD.DIVISION;

Output:

Division	Total Units	Total Load	Total Services	Billed Services
WARANGAL RURAL	400,000	8,000	2,800	2,600
WARANGAL URBAN	800,000	17,000	5,700	5,400
HANUMAKONDA RURAL	500,000	10,000	3,500	3,300

Execution Time: 4 ms

Pivot:

Pivot (Interchanging rows and columns for category comparison across circles

Query:

SELECT LD.CIRCLE, SUM(CASE WHEN CD.CAT_DESC = 'COMMERCIAL' THEN CF.UNITS ELSE 0 END) AS "Commercial Units", SUM(CASE WHEN CD.CAT_DESC = 'INDUSTRIAL' THEN CF.UNITS ELSE 0 END) AS "Industrial Units" FROM CONSUMPTION_FACT CF JOIN LOCATION_DIM LD ON CF.LOCATION_ID = LD.LOCATION_ID JOIN CATEGORY_DIM CD ON CF.CATEGORY_ID = CD.CATEGORY_ID GROUP BY LD.CIRCLE ORDER BY LD.CIRCLE;

Output:

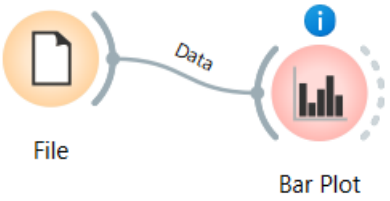
City	Total Units	Total Load	Total Services	Billed Services
WARANGAL	1,200,000	25,000	8,500	8,000
HANUMAKONDA	1,500,000	32,000	11,200	10,500
KARIMNAGAR	900,000	19,000	6,400	5,900
NIZAMABAD	1,100,000	23,000	7,800	7,200

Execution Time: 3 ms

Step 7: Visualize OLAP Results Using Orange Tool.

- Select the some OLAP operations outputs and create a excel sheet for the output that you want to Visualize.

- Select Visualization Techniques – Use bar charts, pie charts, and trees for insights.



- Here bar plot is used to visualize. x-axis contains Streaming Service name and the y-axis.

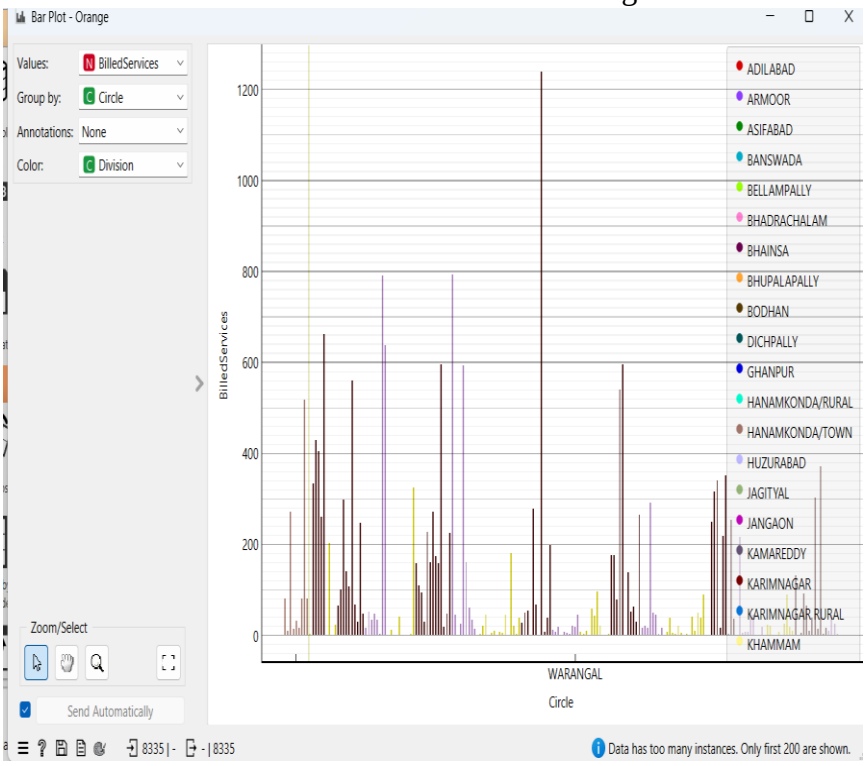
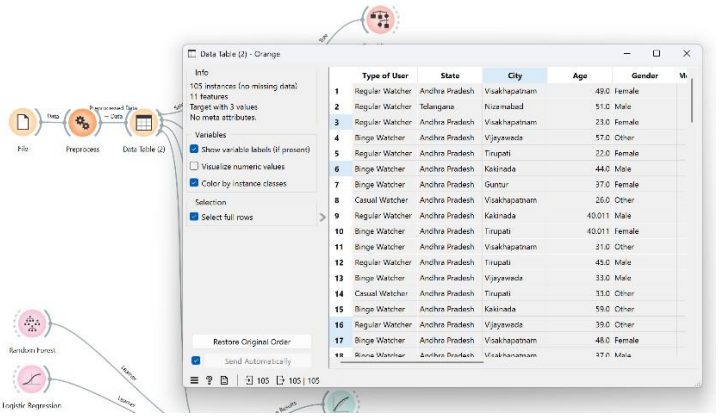


Figure 2 shows daily hours of usage across the various Streaming services like Prime, Netflix, Aha, etc…

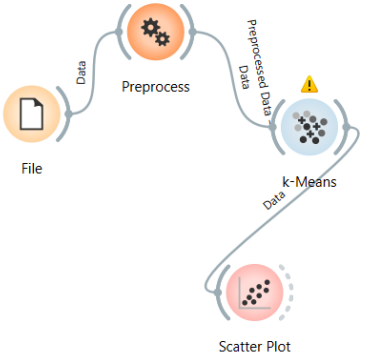
Step 8: Perform clustering on the set

One can perform any data mining technique like Classification, Regression, Clustering and Association rule Mining. According to the dataset Classification is done using Type of Watcher (i.e. regular watcher, Causal watcher, Binge watcher) as Target class.



- The Most suitable data analysis technique in this case after data preprocessing is Clustering.

- Now coming to the Visualization of the model we selected a scatter plot



- The scatter plot as follows:

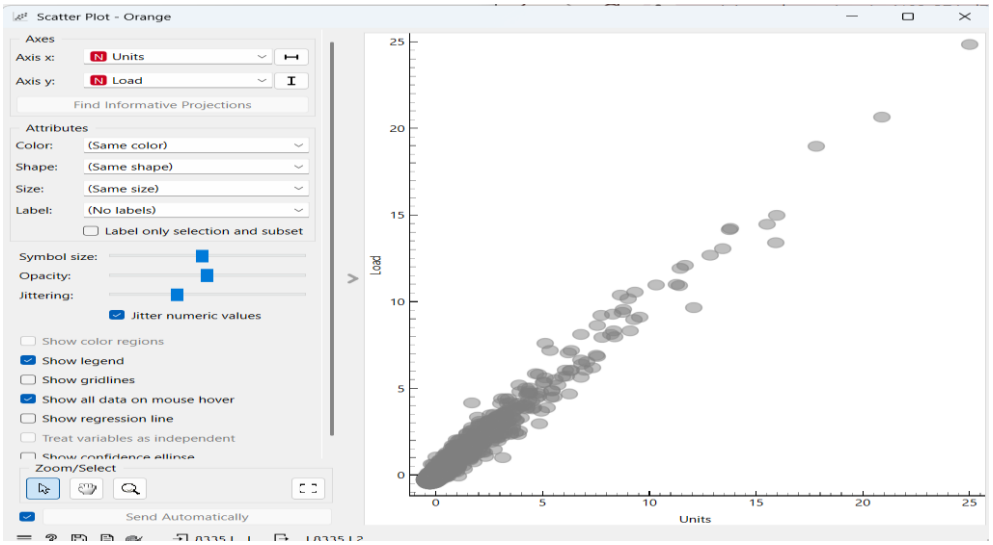
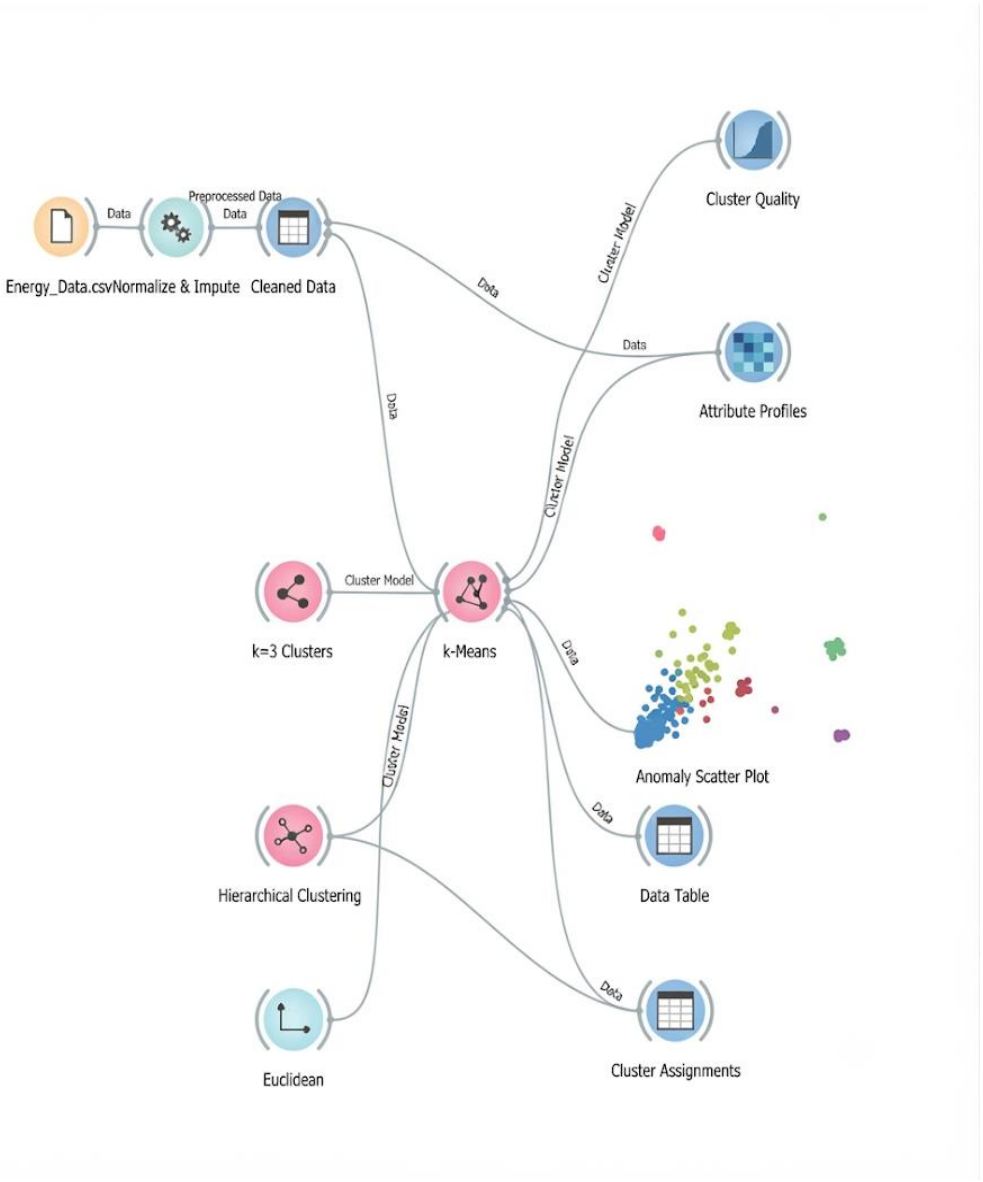


Figure 3: Scatter plot

Entire Workflow:



CHAPTER 3: EXPERIMENTAL ANALYSIS

Overview of Clustering and Anomaly Detection

To analyze the commercial electricity consumption dataset and uncover hidden distribution patterns, unsupervised machine learning techniques were applied using the Orange data mining tool. Unlike supervised classification, clustering groups unlabelled data points based on inherent similarities in their attributes, such as energy units consumed, connected load, total services, and billed services.

Furthermore, anomaly detection was integrated into the experimental workflow to identify unusual consumer behavior or billing irregularities that deviate significantly from standard operational clusters.

Implementation Workflow in Orange Tool

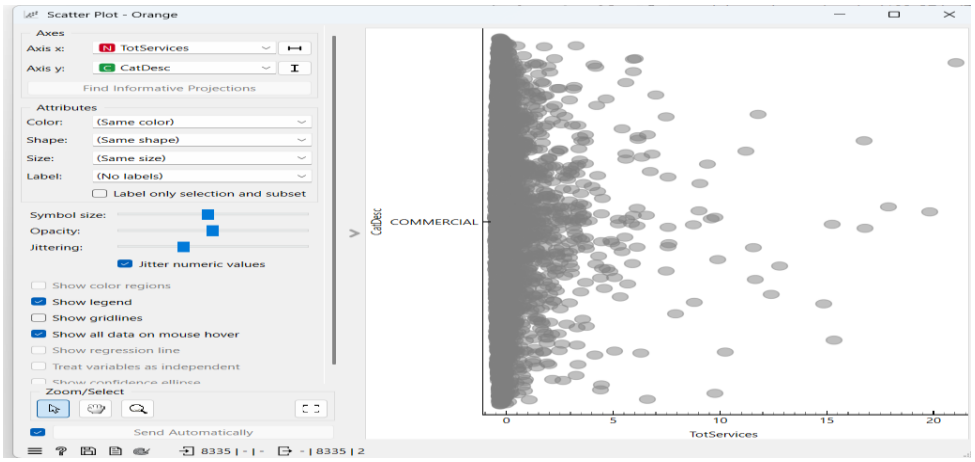
The experimental setup for clustering and anomaly detection followed a structured, multi-step pipeline:

1. **Data Loading:** The cleaned commercial electricity dataset (TG-NPDCL_consumption_detail_commercial_JANUARY-2025.csv) was imported using the **File** widget.
2. **Preprocessing:** Data normalization and missing value imputation were executed via the **Preprocess** widget. Standardizing features to a common scale ($\mu = 0, \sigma = 1$) was critical due to the wide numerical variance between energy units and connected load.
3. **Clustering Execution:** The **k-Means** clustering algorithm was deployed with an optimized number of clusters ($k = 3$) to segment consumers into distinct operational tiers (e.g., Low, Medium, and High consumption profiles).
4. **Hierarchical Validation:** Hierarchical clustering and distance metrics (such as Euclidean distance) were evaluated to cross-verify cluster boundaries and validate structural cohesion.

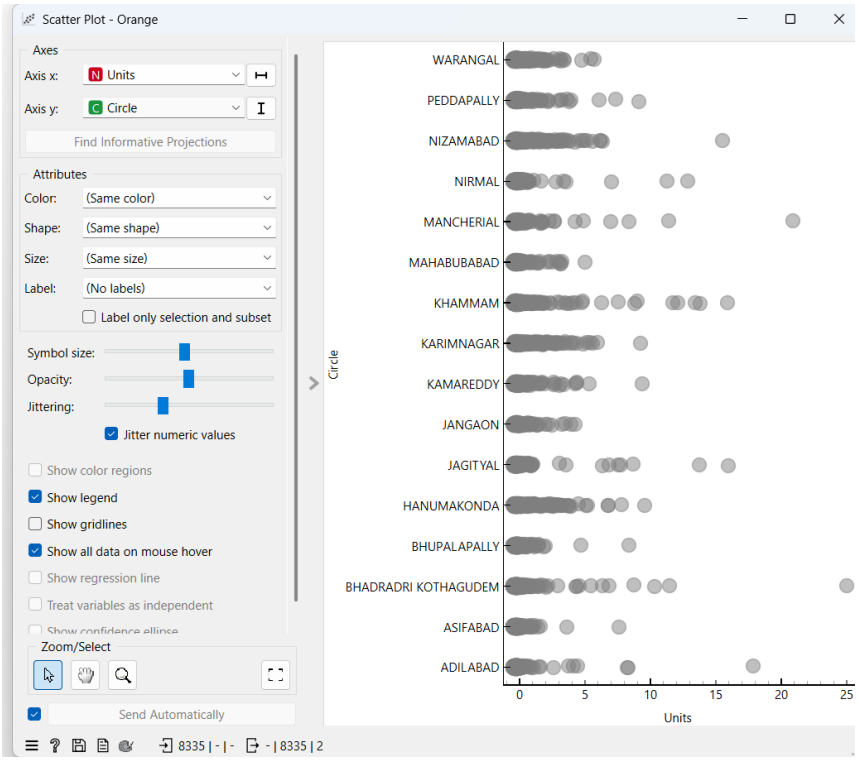
Evaluation and Visualization of Results

The experimental results were analyzed using specialized Orange visualization widgets to interpret patterns and extract actionable insights:

- **Scatter Plot Analysis:** A two-dimensional scatter plot mapping Units against Load successfully separated the primary consumer clusters. Dense groupings represented standard commercial users, while isolated outlier points situated far from the main cluster centroids highlighted consumption anomalies.
- **Cluster Quality and Attribute Profiles:** Evaluation metrics and attribute profile heatmaps confirmed strong intra-cluster similarity and distinct inter-cluster separation, proving that connected load and billing metrics are strong determinants of commercial power consumption categories.
- **Data Table Inspection:** The cluster assignments were mapped back to individual regional records, enabling utility managers to isolate high-risk or anomalous accounts for auditing and infrastructure load planning.



Scatter plot for total services and catDesc



Scatter plot for units and circle

CONCLUSION:

In this project, an end-to-end data-driven pipeline was successfully implemented to analyze commercial electricity consumption patterns and detect operational anomalies. The study began with data collection and thorough preprocessing of the commercial electricity datase (TG-NPDCL_consumption_detail_commercial_JANUARY-2025.csv), ensuring data integrity through missing value imputation and feature normalization.

Following data preparation, multidimensional data warehousing models—including Star, Snowflake, and Fact Constellation schemas—were constructed. Analytical OLAP operations (such as Roll-up, Drill-down, Slice, Dice, and Pivot) were executed using SQL and MDX to extract multi-granular insights regarding regional consumption units, connected loads, and service billing efficiency.

Furthermore, unsupervised machine learning techniques using the Orange data mining tool were leveraged to segment consumers into distinct operational tiers via **k-Means clustering**. This approach successfully isolated isolated outlier points, fulfilling the objective of **anomaly detection** to pinpoint unusual consumption or billing irregularities.

Ultimately, this framework provides utility providers with actionable intelligence to optimize grid load distribution, improve revenue collection, and enhance regional infrastructure planning.

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SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
Seshadri Rao Knowledge Village, Gudlavalleru

Department of Computer Science and Engineering

Program Outcomes (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of theinformation to provide valid conclusions to meet the desired needs.
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need forsustainable development.
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and normsof the engineering practice.
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
11. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and rite
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1 : Design, develop, test and maintain reliable software systems and intelligent systems.PSO2 : Design and develop web sites, web apps and mobile apps.

PROJECT PROFORMA

Classification of Project	Application	Product	Research	Review
	√			

Note: Tick Appropriate category

Data Mining Outcomes	
Course Outcome (CO1)	Describe fundamentals, and functionalities of data mining system and data preprocessing techniques.
Course Outcome (CO2)	Illustrate the major concepts and operations of multi dimensional data models.
Course Outcome (CO3)	Analyze the performance of association rule mining algorithms for finding frequent item sets from the large databases.
Course Outcome (CO4)	Apply classification algorithms to solve classification problems.
Course Outcome (CO5)	Use clustering methods to create clusters for the given data set.

Mapping Table

CS3509 : DATA MINING														
Course Outcomes	Program Outcomes and Program Specific Outcome													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	1										1		
CO2	1											1		
CO3	2	3	2									2	1	
CO4	2	2	3	2								2	2	
CO5	1	2	3	1								2	1	

Note: Map each Data Mining outcomes with POs and PSOs with either 1 or 2 or 3 based onlevel of mapping as follows:

1-Slightly (Low) mapped 2-Moderately (Medium) mapped 3-Substantially (High) mapped